Robustness of the SHiP muon shield to parameter tolerances

October 24, 2026

Abstract

How much do the muon hits on the SHiP sensitive plane change when the 43 free parameters of the optimised muon shield deviate from their nominal values? A neural surrogate of the per-muon hit probability (an ensemble of four DeepONets) gives the gradient, the Hessian and candidate worst-case directions, and every conclusion is checked with the full simulation on 10^9 muons. The nominal design gives about 830 hits per 10^9 muons. If every parameter may be off by up to ε, the worst design found within these tolerances (searched on the surrogate, then simulated) multiplies the hits by 1.72 ± 0.07 at $\varepsilon = 1\%$, 3.8 ± 0.2 at 2%, 9.0 ± 0.3 at 3%, 132 ± 5 at 5% and 3440 ± 120 at 10%. A single parameter at its $\pm10\%$ limit gives at most $\times2.1\pm0.1$ (the field of M7). Random independent deviations are much milder: harmless at $\pm2\%$, mostly harmless at $\pm5\%$, with a heavy tail at $\pm10\%$. Bounding the worst case takes sub-percent tolerances on a few parameters, mostly of M7 and M2: the allocation found for at most $\times2$ (simulated: $\times1.56$) holds M7's field, M7's length and M2's entrance width to $\pm0.5\%$. Near the nominal design every harmful direction acts through muons of 30–75 GeV, while the muons above 150 GeV respond the opposite way.

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1 Setup

The shield. The design is the optimised configuration (`try_opt`) of seven magnets M1–M7 (M0 is fixed), with 43 free parameters (Table 10): lengths `Z_len` and gaps `Z_gap`, the transverse sizes `dXIn`, `dXOut`, the inner gaps `gapIn`, `gapOut`, the yoke widths `dXYokeIn/Out` (all in cm) and the field B of each magnet (the `NI` column, the target field in tesla). Fields are uniform within each magnet block; the cavern is included.

Tolerances. Each parameter may vary by ±10% of its nominal value:

$$\phi_i = \phi_{0,i}\,(1 + 0.1\,u_i), \qquad u_i \in [-1, 1]. \tag{1}$$

$u = 0$ is the nominal design and $u_i = \pm 1$ puts parameter i alone at its limit. A step of $\|u\| = 1$ is "one tolerance unit": as large as one parameter at its limit, but spread over all of them (RMS change 1.5% per parameter). The corners of the box are at $\|u\| = \sqrt{43} = 6.6$. M6 and M7 have negative fields, so a negative change of their B lowers $|B|$.

Muons and figure of merit. Muons are drawn from the generative flow of `generator_muons` (the physical flux) and placed on axis at $z = -1$. The figure of merit H is the number of muons crossing the last sensitive plane per 10^9 muons. At the nominal design, eight simulator runs on the muons of the analysis give 803–852 (each ±29; the transport is stochastic) and independent muons 820 ± 14, a hit rate of 8×10^{-7}; the surrogate gives 810. Every simulator number uses 10^9 muons, the same ones for all designs, with Poisson errors (the transport is stochastic, so this holds even on shared muons).

2 The surrogate

Model. The surrogate (LCSO) models the hit probability of each muon x for a design u,

$$p(\text{hit} \mid x, u) = \sigma\big(b(u) \cdot t(x) + c\big), \tag{2}$$

a DeepONet with a branch network b for the design and a trunk network t for the muon (p_x, p_y, p_z, charge). The hits $H(u)$ are the integral of p over the muons, a smooth function of u with exact derivatives. Four DeepONets (two hidden layers of 128 units, 32 features) are trained with different seeds and their logits averaged; the average is itself a DeepONet with concatenated features. A seed whose final validation loss is more than 10% above the median of its siblings is replaced by a new one (this happened once: 0.0179 against 0.0152–0.0156); the rule uses only the held-out training designs, not the references. Errors on surrogate quantities are the standard error of the four members: the surrogate's own seed-to-seed uncertainty.

Training data. The training data are simulated once, in eight pools (Table 1), and reused for every epoch; nothing is re-simulated during training. Each pool is a set of designs, each simulated with its own fresh muons from the flow:

- *Designs.* $u = \delta\, s\, r$, with s a scrambled Sobol point in $[-1, 1]^{43}$ and r uniform in $(0, 1)$, for a region of half-width δ ($\delta = 1$: the ±10% box). The radial factor r concentrates the designs towards the nominal one: of the 13 947 designs drawn this way, 4475 have every parameter within ±3%. The first 16 designs of each base pool are the nominal design itself (8×10^8 muons at $u = 0$ in all).
- *Far from the nominal design.* The surrogate was weakest there (Table 2), so two pools were added. The *uniform* pool drops the radial factor: 1024 Sobol designs uniform in the box, all with some parameter beyond ±8.5% (two crashed the simulator and are missing). The *adaptive* pool (212 designs) targets where the conclusions rest and where the surrogate was

least sure: the worst designs found within ±1–10% and those of the tolerance allocations (Sec. 4.5), each with four points on its line from the nominal design and 12 perturbations within its box (24 for the one the simulator showed mispredicted, at ±10%), and the 64 of 12 288 random designs on which the members of a first ensemble, trained on the other six pools, disagreed most.

- *Muons.* Drawn from the flow at a raised *momentum* temperature T (only the momentum part of the flow is tempered), which broadens the momentum spectrum and raises the hit rate from 8×10^{-7} (the physical flux) to 1.4×10^{-4} ($T = 1.5$) and 1.8×10^{-3} ($T = 2$). This does not bias the model, which learns $p(\text{hit} \mid x, u)$ per muon; the temperature only decides where in momentum the training data is dense. The $T = 1$ pool, at the physical flux, holds only 6900 hits.

- *Rows.* Every hit and 0.4% of the misses are kept, a case-control sample; the logit is fitted with the offset $\log 250$, so the model remains one of the full probability.

- *Split.* The designs, not the rows, are split: 13 663 for training and 1518 held out for early stopping. The validation references of Sec. 2.1 are separate simulations, never used for training.

In all, 6.2×10^{10} simulated muons give 3.4×10^7 hits and 2.8×10^8 rows. At the physical flux, as many hits would need 4×10^{13} muons, i.e. about 85 000 designs of 5×10^8 muons; the same number of simulated muons would buy only ~ 120 such designs.

Table 1: Training pools. Designs are drawn in the region given, pulled towards the nominal design except in the uniform and adaptive pools (see text); designs that crashed the simulator are missing.

pool	designs	muons per design	momentum T	region	hits
main	2048	10^7	1.5	±10%	3.0×10^6
small region	512	10^7	1.5	±5%	7.0×10^5
large region	363	10^7	1.5	±20%	7.6×10^5
cold	512	10^7	1	±10%	6.9×10^3
hot	512	10^7	2	±10%	9.2×10^6
many designs	10 000	10^6	2	±10%	1.8×10^7
uniform	1022	10^7	1.5	±10%, uniform	1.8×10^6
adaptive	212	10^7	1.5	selected	4.0×10^5
total	15 181	6.2×10^{10}			3.4×10^7

Integration over the muons. With the muons on axis, a muon is its momentum, azimuth and charge, which are independent in the flow: (p_z, p_t) is the image of a 2-d standard normal latent. The surrogate's hits, gradient, Hessian and per-momentum splits are therefore computed on a quadrature grid (latent × azimuth × charge, App. B) instead of by sampling muons: exact to 10^{-5} in seconds, where 10^9 sampled muons leave 2–5% of noise. Only the simulator uses sampled muons.

2.1 Validation against the simulator

Reference designs were simulated on stored muons, on which the surrogate also predicts: 10^9 muons of the physical flux (the nominal design four times, 16 designs in the half box $|u_i| \le \frac{1}{2}$, 16 in the box), and a high-momentum check sample of 4×10^8 muons drawn like the $T = 1.5$ pools ($\sim 5 \times 10^4$ hits per design, so precise enough for one-at-a-time effects: the same designs plus every parameter alone at ±10%). Table 2 and Fig. 1 compare the surrogate with the simulator on them.

The one-at-a-time effects are reproduced to 3.9%, the designs in the half box to 4%. The weak point is far from the nominal design (13% in the box): there the hits come mostly from muons of 30–75 GeV and below, which are rare in every training sample, while the high-momentum hits that

Table 2: Surrogate vs. simulator on the reference designs: median $|\log(\text{pred}/\text{sim})|$ (signed for the nominal design), and the relative vector errors of the 43 one-at-a-time (OAT) secants $\frac{1}{2}[H(+e_i) - H(-e_i)]$ and curvatures $H(+e_i) + H(-e_i) - 2H_0$ on the check sample. "Noise": what a perfect model would score, from the simulator's Poisson noise.

	noise	surrogate
Nominal design	1.7%	−2.0%
Designs in the half box	2.1%	4.1%
Designs in the box	1.6%	13%
Check sample: nominal design	0.2%	+0.6%
Check sample: designs in the box	0.25%	3.9%
OAT secants (cosine)	1.6%	3.9% (0.999)
OAT curvatures (cosine)	7.6%	12% (0.993)

dominate the training data are predicted to a few percent. Within one tolerance unit, where the derivatives are read, the walks of Sec. 4.3 agree with the simulator to 1–2σ.

The derivatives are well determined: the four members' gradients at $u = 0$ agree to cos = 0.974–0.992 and their top Hessian eigenvectors to 0.997–0.999 with the ensemble's. The second and third eigenvectors are nearly degenerate and not individually determined. How the architecture and training were chosen is summarised in App. C.

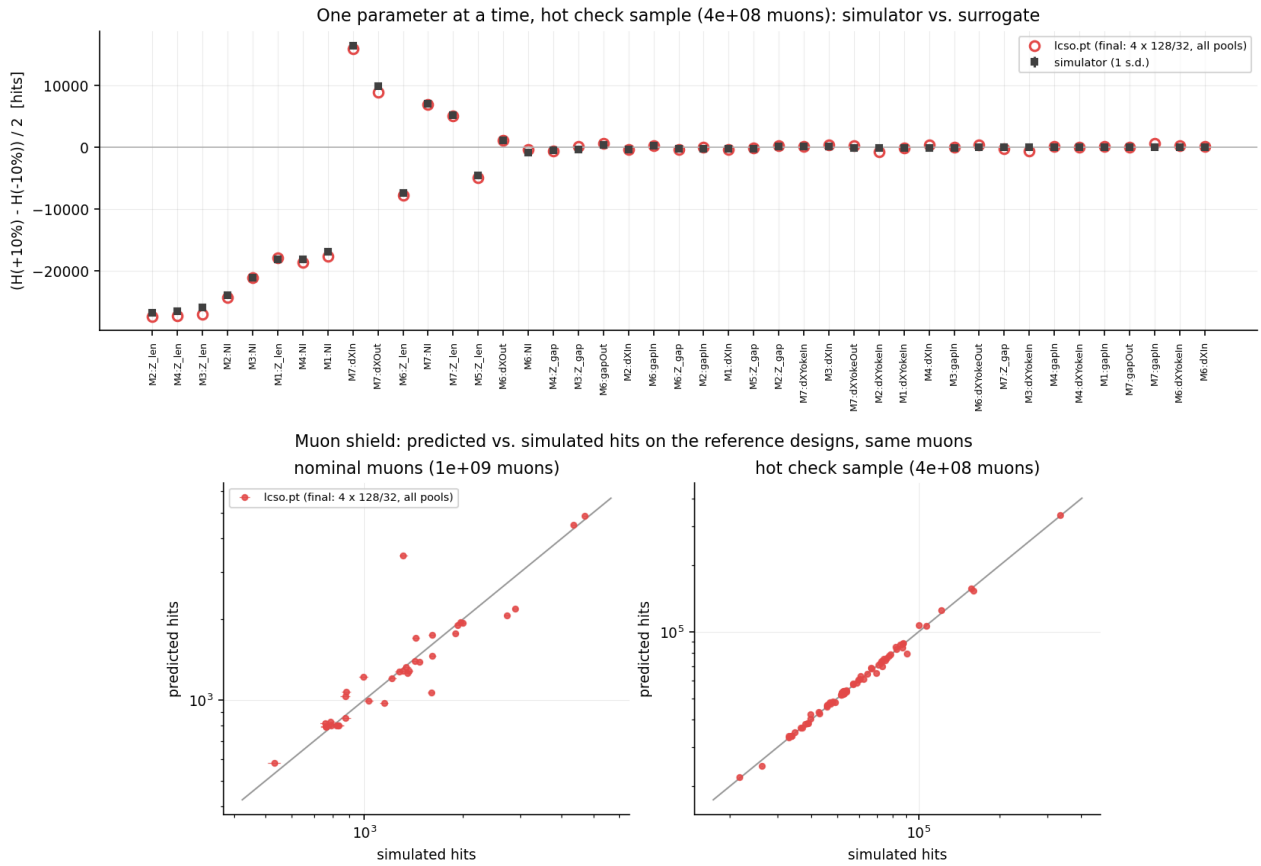


Figure 1: Top: every parameter alone at $\pm 10\%$, secant $\frac{1}{2}[H(+10\%) - H(-10\%)]$ on the check sample: simulator (squares) and surrogate (circles). Bottom: predicted vs. simulated hits of the reference designs.

3 Method

From the surrogate we compute

- the gradient $g = \nabla_u H$ and Hessian $\nabla_u^2 H$ at $u = 0$, and the exact one-at-a-time change $H(\pm e_i)/H_0 - 1$ of each parameter at its limit;
- the active subspace, the eigenvectors of $C = \mathbb{E}_x[\nabla_u p \, \nabla_u p^\top]$: the directions in which the hit probability of individual muons changes most;
- candidate worst directions, each of unit length: `ascent` ($v \propto g$), `worst` (maximiser of the quadratic model $g \cdot v + \frac{1}{2} v^\top \nabla^2 H \, v$ on $\|v\| = 1$), `hess1` (top Hessian eigenvector), `act1`, `act2` (top active directions), `corner` ($v \propto \operatorname{sign} g$: every parameter moved by the same amount) and `random`;
- the worst corner of the box, by projected-gradient ascent (`pga`);
- 4096 random designs in the box;
- the worst design for a given tolerance and a tolerance allocation (Sec. 4.5).

Each direction is then walked, $u = t\,v$ for $t \in \{-1, -\frac{1}{2}, \frac{1}{2}, 1\}$ (for `pga`, up to the box face), and simulated.

Table 3: The most sensitive parameters (surrogate): first-order sensitivity g_i/H_0 per full tolerance, and the exact change at ±10% (errors: the members' standard error).

parameter	nominal	g_i/H_0	$H/H_0 - 1$ at +10%	at −10%
M7:B	−1.9 T	−0.494	-0.26 ± 0.01	$+1.03 \pm 0.02$
M2:`dXIn`	68.04 cm	−0.384	$+0.17 \pm 0.02$	$+0.99 \pm 0.03$
M7:`Z_len`	193.1 cm	−0.389	-0.23 ± 0.01	$+0.69 \pm 0.03$
M7:`dXYokeIn`	124.8 cm	+0.403	$+0.47 \pm 0.03$	$+0.02 \pm 0.03$
M3:`dXYokeIn`	85.88 cm	−0.315	-0.11 ± 0.01	$+0.45 \pm 0.04$
M7:`dXOut`	67.81 cm	−0.221	-0.11 ± 0.02	$+0.41 \pm 0.03$
M2:B	1.9 T	−0.209	-0.14 ± 0.02	$+0.34 \pm 0.02$
M6:`Z_len`	168.0 cm	−0.240	-0.19 ± 0.01	$+0.30 \pm 0.01$
M7:`dXYokeOut`	94.92 cm	+0.251	$+0.27 \pm 0.02$	-0.09 ± 0.01
M2:`Z_len`	224.9 cm	−0.040	$+0.10 \pm 0.03$	$+0.25 \pm 0.02$
M4:`Z_len`	224.0 cm	−0.127	-0.07 ± 0.02	$+0.25 \pm 0.01$
M3:B	1.9 T	+0.059	$+0.24 \pm 0.03$	$+0.13 \pm 0.02$
M3:`Z_len`	225.0 cm	−0.013	$+0.14 \pm 0.02$	$+0.22 \pm 0.02$
M1:B	1.9 T	−0.149	-0.10 ± 0.01	$+0.22 \pm 0.01$
M3:`dXIn`	64.81 cm	+0.110	$+0.21 \pm 0.03$	-0.02 ± 0.01

4 Results

4.1 Single parameters

32 of the 43 one-at-a-time effects are resolved beyond three times the members' scatter (Fig. 2, Table 3). The most sensitive parameter is the *field of M7*: lowering $|B|$ by 10% doubles the hits. Next come the entrance width of M2 (narrowed by 10%: also ×2), the length, yoke widths and exit width of M7, the inner yoke of M3, the field of M2 and the length of M6. Most responses are one-sided: one side of the tolerance hardly matters, the other is harmful. A few (the lengths of M2 and M3, the field of M3) are harmful on both sides, i.e. the nominal value sits at their minimum. Per magnet (Fig. 3), M7 carries by far the largest share of the gradient (77% of its norm), then M2 (42%), M3 (32%) and M6 (27%).

Simulated. The eight largest were simulated at their worse limit (Table 6): all eight rise significantly, six within 1–2σ of the surrogate; the exit width of M7 rises more than predicted ($\times 1.85 \pm 0.08$ vs. 1.41) and the inner yoke of M3 less (1.26 ± 0.06 vs. 1.45). Their scans were also simulated at ±5% and ±10% (Fig. 4): 25 of the 32 points agree within 2σ (14 of 16 at ±5%). The misses are the downstream geometry of M7 narrowed, where the surrogate is too smooth (exit width at −5% and −10%: 1.36 and 1.91 simulated vs. 1.15 and 1.41; inner yoke at −10%: 1.25 vs. 1.02), and the entrance width of M2 widened by 10%, harmless in the simulator (1.01 ± 0.05) where the surrogate predicts a small rise (1.17). The inner yoke of M7 is harmful on both sides in the simulator.

4.2 Curvature and active subspace

One Hessian eigenvalue dominates (Fig. 5): $3.41 \pm 0.13\, H_0$ per unit^2, then 1.01, 0.86 and $\lesssim 0.17\, H_0$; 14 are negative, none below $-0.18\, H_0$. Along the gradient the second-order term already exceeds the first-order one at one tolerance unit ($1.19\, H_0$ vs. $1.07\, H_0$): the response is strongly non-linear at that scale.

The active subspace is effectively one-dimensional (98.8% of $\mathbb{E}[\|\nabla_u p\|^2]$). Its direction `act1` shortens M1–M4 and lowers their fields together, i.e. it changes the upstream bending power (Table 5). It is almost orthogonal to the gradient (cos = 0.06): individual muons are deflected most along it, but the changes largely cancel in the total.

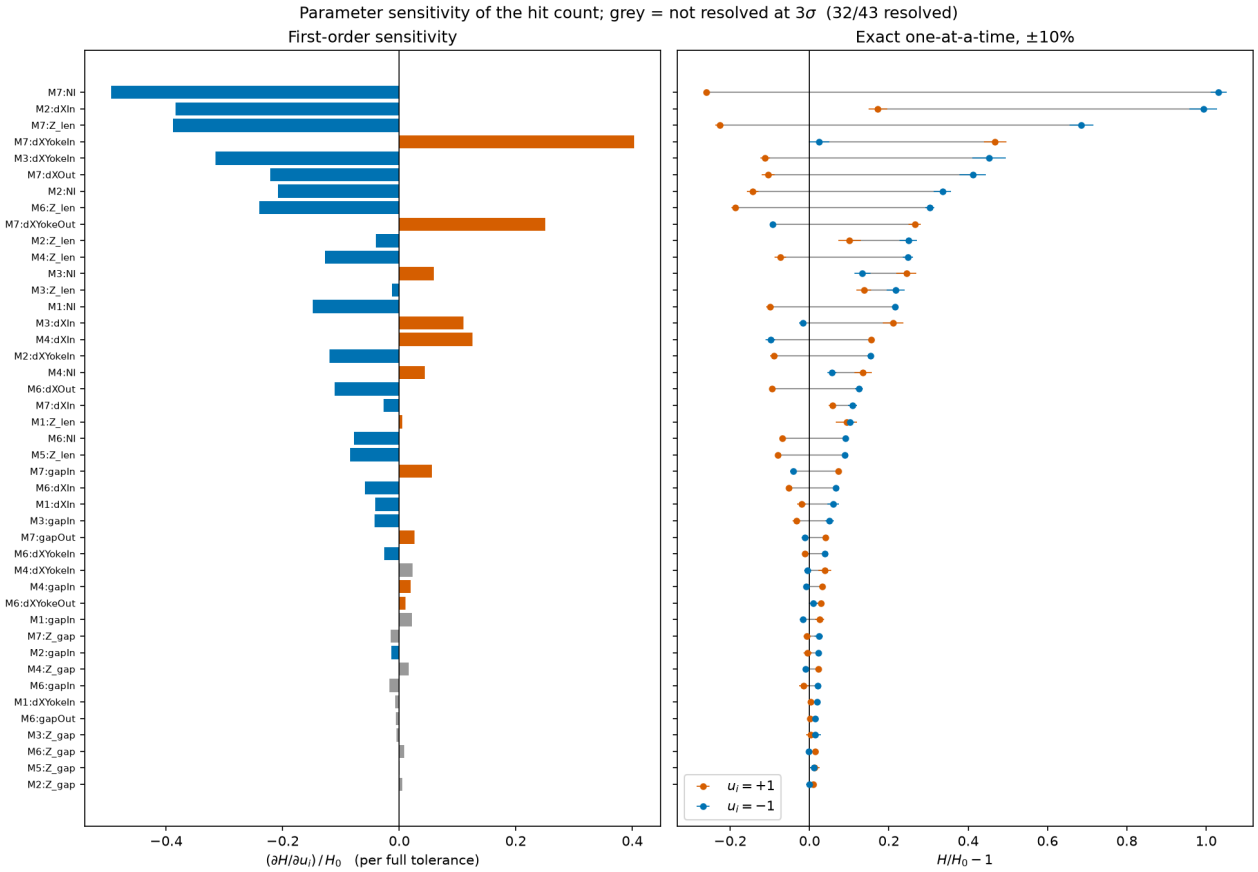


Figure 2: Sensitivity of each parameter. Left: gradient per full tolerance. Right: exact change with the parameter at +10% (orange) and −10% (blue).

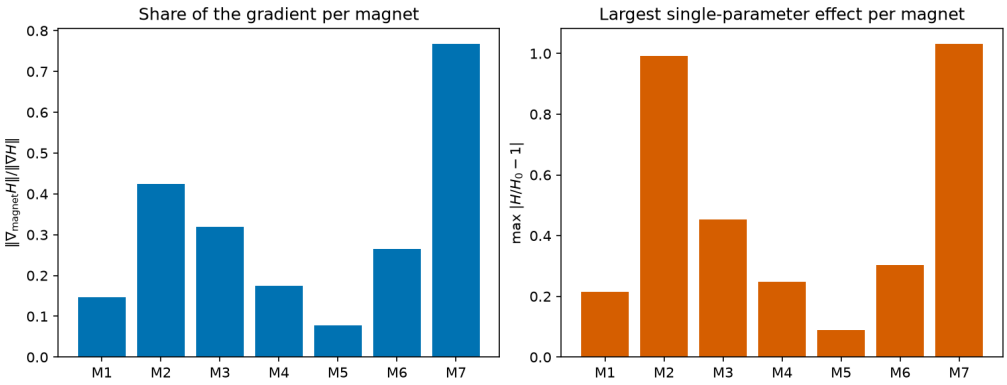


Figure 3: Per magnet: share of the gradient norm and largest single-parameter effect.

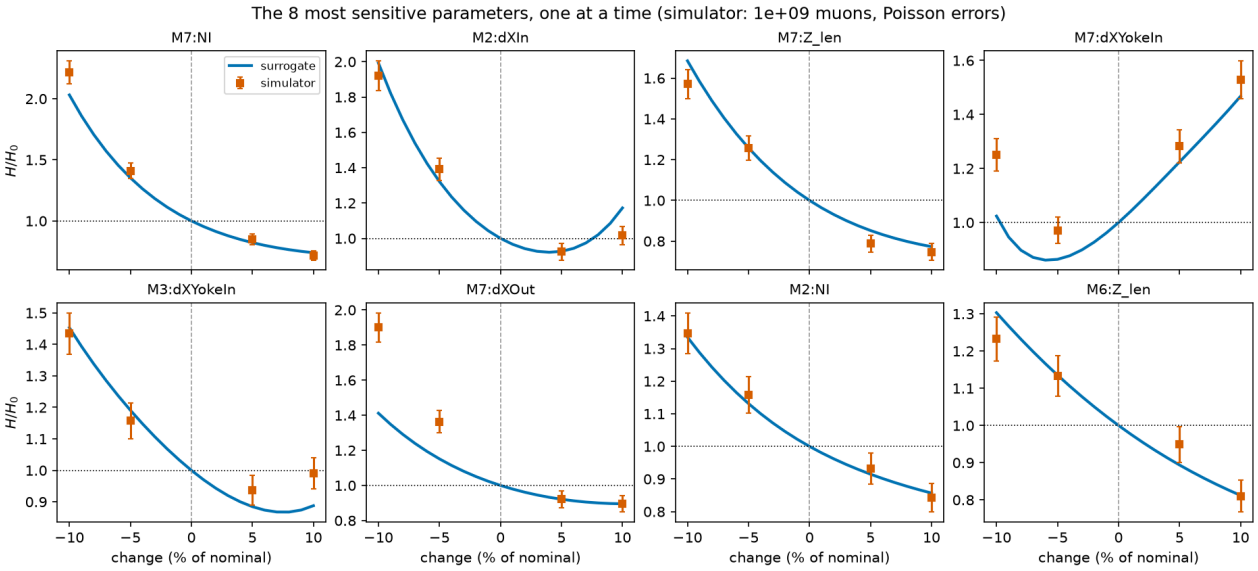


Figure 4: One-at-a-time scans of the eight most sensitive parameters: surrogate (blue) and simulator at ±5% and ±10% (orange).

4.3 Worst directions

Small coordinated changes are far more harmful than a large single one. Moving every parameter by at most 4.3% (RMS 1.5%) along `worst` multiplies the hits by 4.95 ± 0.19 (Table 4). The simplest such pattern, `corner`, moves *every* parameter by exactly 1.5% with the sign of its gradient and doubles the hits. The worst single parameter needs a 10% error for a similar effect. A random direction of the same size changes little (0.87 ± 0.04): the damage comes from the pattern of signs, not the size of the step.

The surrogate gets both the directions and the sizes. Its two best directions (`worst` and `hess1`, $\cos = 0.98$) are the two the simulator ranks highest, then `ascent`. Within one tolerance unit it agrees with the simulator to 1–2σ almost everywhere (Fig. 7); the exceptions are `act2` at $t = 1$ (1.20 vs. 1.39 ± 0.06) and the $t = -1$ ends of `ascent` and `worst`, where it is 50% and 17% high.

The quadratic model is not enough. Built from the gradient and Hessian, it underestimates every harmful direction (3.6 instead of 4.95 along `worst`, Fig. 6): the hits grow roughly exponentially. The derivatives give the directions; magnitudes come from the surrogate and the simulator.

The active subspace is a direction of curvature, not of slope. `act1` gives $\times 2.44$ at $t = +1$ and also $\times 1.50$ at $t = -1$: shortening or lengthening magnets 1–4 together, with their fields, is

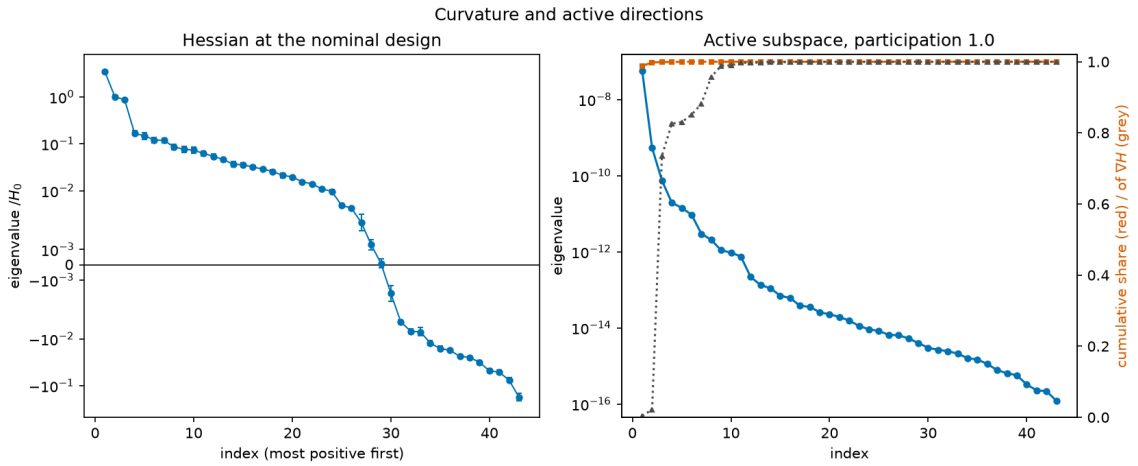


Figure 5: Left: Hessian eigenvalues in units of H_0. Right: active-subspace eigenvalues (blue), their cumulative share (red) and the fraction of the gradient they contain (grey).

Table 4: Hits one tolerance unit along each direction, surrogate and simulator (nominal: 843 simulated hits). "max / RMS": the largest and the RMS change over the 43 parameters.

direction	max / RMS change	surrogate H/H_0	simulator H/H_0	sim. hits	at $t = -1$ (sim.)
`worst`	4.3% / 1.5%	5.20 ± 0.17	4.95 ± 0.19	4174	1.00 ± 0.05
`hess1`	4.2% / 1.5%	5.26 ± 0.18	5.19 ± 0.20	4377	1.31 ± 0.06
`ascent`	4.6% / 1.5%	3.86 ± 0.06	4.02 ± 0.15	3389	0.60 ± 0.03
`act1`	3.9% / 1.5%	2.56 ± 0.07	2.44 ± 0.10	2059	1.50 ± 0.07
`corner`	1.5% / 1.5%	2.15 ± 0.04	2.01 ± 0.08	1690	0.67 ± 0.04
`act2`	9.4% / 1.5%	1.20 ± 0.03	1.39 ± 0.06	1174	0.87 ± 0.04
`random`	— / 1.5%	0.96 ± 0.00	0.87 ± 0.04	731	1.10 ± 0.05

harmful either way, and the nominal design sits at the balance point (Sec. 4.8). In the plane of `worst` and `act1` (Fig. 8) the nominal design lies in a shallow valley.

4.4 Concrete designs along the worst direction

Table 6 lists concrete designs, all simulated. "Worst, max ε" is the worst direction scaled so that no parameter moves by more than ε. It lies within the $\pm\varepsilon$ tolerances, so it bounds their worst case from below; most of its parameters stay well inside them, and the worst design within the tolerances is much worse (Sec. 4.5).

Exponential growth along the direction. Along the worst direction (Fig. 9) the hits grow by $\times 1.24$ at a maximum change of 1%, $\times 1.8$ at 2%, $\times 2.7$ at 3%, $\times 6.8$ at 5% and $\times 73$ at 10%: a factor e per $\sim 2.3\%$ of tolerance. The surrogate follows to within 10% up to 5% and is 25% low at 10%. The opposite direction is harmless at 5% ($\times 1.2$) but not at 10% ($\times 5.2$), through other muons (Sec. 4.8).

The rise is collective. The largest component alone (M2:`dXIn` -10%) gives $\times 1.9$; the top three (adding a weaker and shorter M7) $\times 6.9$; the top five $\times 13$; the top ten $\times 34$; all 43 $\times 73$ (Fig. 10). Each parameter contributes a moderate factor, and the factors multiply.

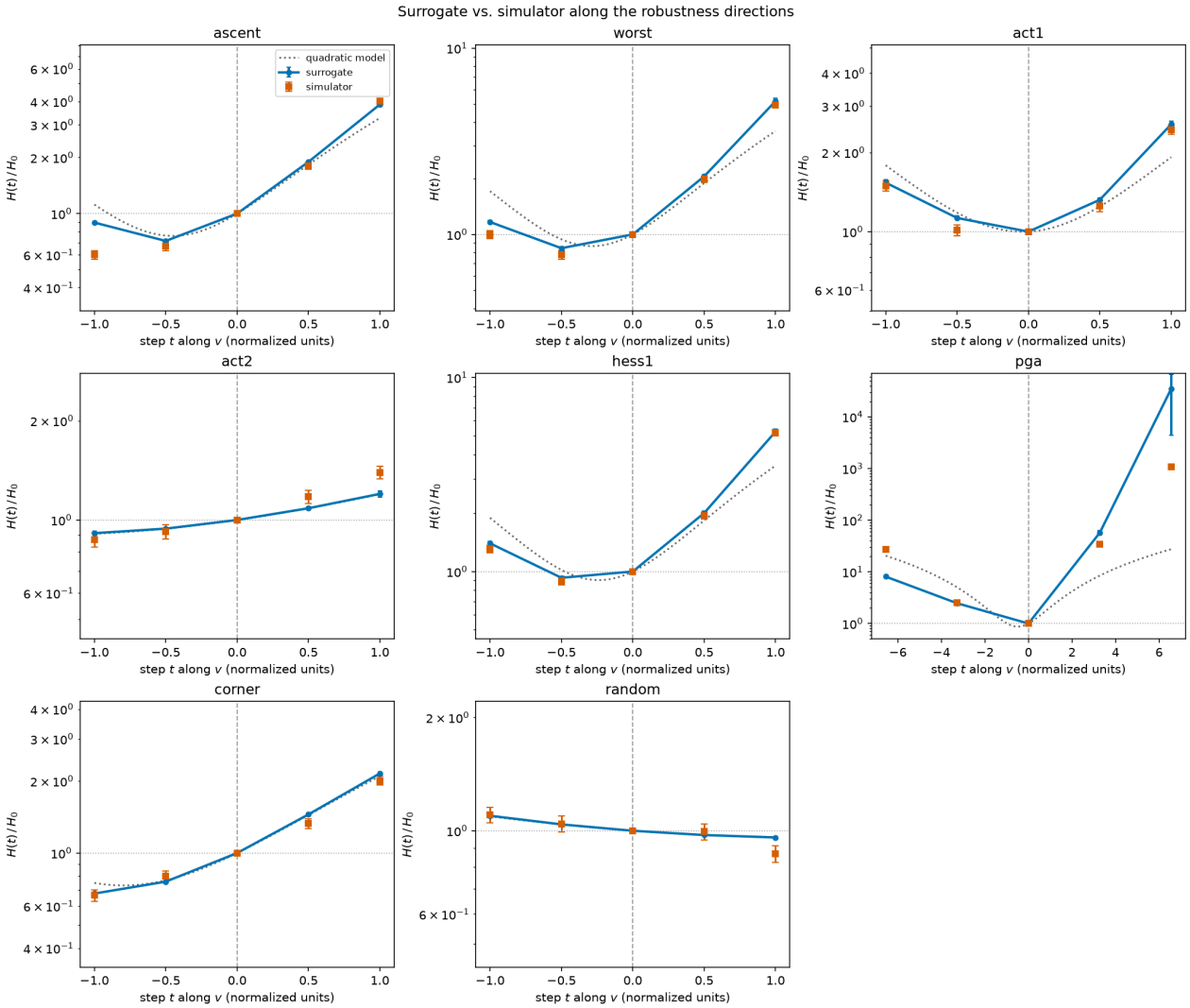


Figure 6: Walks along each direction: surrogate (blue; error bars: the members), simulator (orange) and the quadratic model (dotted).

Table 5: What each direction changes at one tolerance unit: its largest parameter changes, relative and in physical units (cm, or T for B).

direction	largest changes	in words
`worst`	M2:`dXIn` −4.3% (−2.9), M7:B −4.2% ($\lvert B\rvert$ −0.08 T), M7:`Z_len` −3.5% (−6.8), M7:`dXYokeIn` +2.9% (+3.7), M3:`dXYokeIn` −2.9% (−2.5), M3:B +2.3% (+0.04 T), M7:`dXOut` −2.3% (−1.5)	weaker and shorter M7 with wider yokes, narrower M2 entrance
`hess1`	M2:`dXIn` −4.2% (−2.9), M7:B −3.7% ($\lvert B\rvert$ −0.07 T), M7:`Z_len` −3.2% (−6.3), M3:B +2.9% (+0.06 T), M3:`dXYokeIn` −2.6% (−2.2), M2:`Z_len` +2.5% (+5.6), M3:`Z_len` +2.4% (+5.3)	as `worst`
`ascent`	M7:B −4.6% ($\lvert B\rvert$ −0.09 T), M7:`dXYokeIn` +3.8% (+4.7), M7:`Z_len` −3.6% (−7.0), M2:`dXIn` −3.6% (−2.4), M3:`dXYokeIn` −2.9% (−2.5), M7:`dXYokeOut` +2.3% (+2.2), M6:`Z_len` −2.2% (−3.8)	the gradient: M7 weaker and shorter
`corner`	all 43 parameters by exactly ±1.5%, each with the sign of its gradient	the simplest uniform pattern
`act1`	M2, M3, M4:`Z_len` −3.9% each (−8.7 to −8.8), M2:B −3.5%, M3:B −3.1%, M4:B −2.8% (−0.05 to −0.07 T), M1:`Z_len` −2.7% (−4.0)	magnets 1–4 shorter and weaker: less bending upstream
`act2`	M7:`dXOut` −9.4% (−6.4), M7:`Z_len` +2.5% (+4.8), the rest at most 1%	the exit width of M7

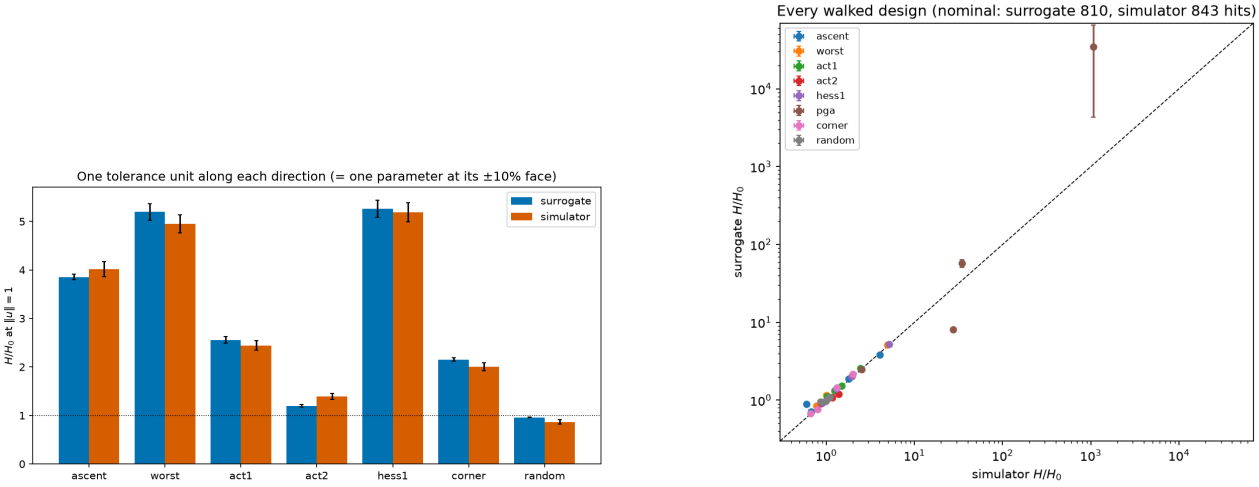


Figure 7: Left: H/H_0 at one tolerance unit along each direction. Right: surrogate vs. simulator for all 33 walked designs; the outliers are the box designs of Sec. 4.5.

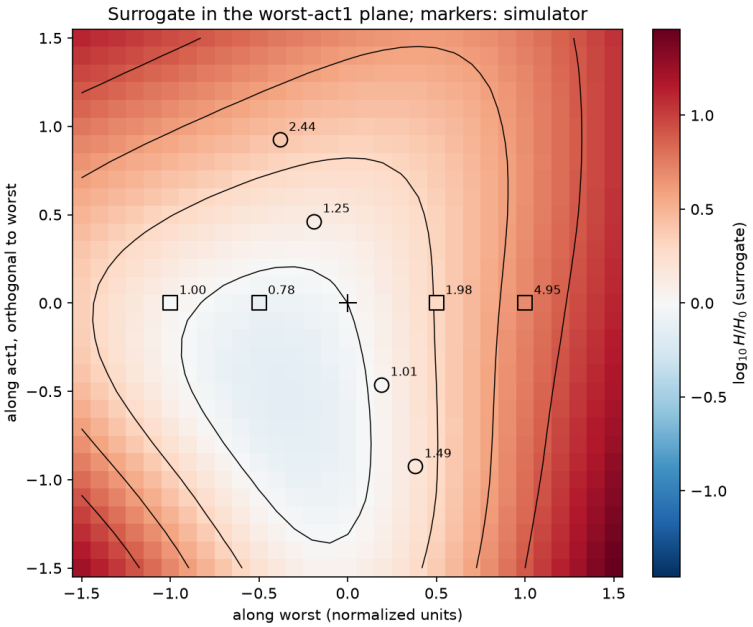


Figure 8: The surrogate in the plane of `worst` (x) and the part of `act1` orthogonal to it (y); markers: simulated designs with their H/H_0.

Table 6: Concrete designs, simulated (nominal 847 ± 29 hits) and predicted. At $\varepsilon = 10\%$ the worst direction changes M2:`dXIn` by -6.8 cm, $|B|$ of M7 from 1.9 to 1.71 T, M7:`Z_len` by -15.9 cm, M7:`dXYokeIn` by $+8.5$ cm, M3:`dXYokeIn` by -5.7 cm, M3:B by $+0.10$ T, M7:`dXOut` by -3.6 cm, M7:`dXYokeOut` by $+4.1$ cm, M3:`dXIn` by $+2.6$ cm, M2:`Z_len` by $+8.9$ cm, and 33 more parameters by less than 4%.

design	what changes	hits	sim. H/H_0	surr. H/H_0
worst, max 1%	all 43, RMS 0.35%	1047	1.24 ± 0.06	1.32
worst, max 2%	all 43, RMS 0.7%	1499	1.77 ± 0.08	1.93
worst, max 3%	all 43, RMS 1.05%	2276	2.69 ± 0.11	2.95
worst, max 5%	all 43, RMS 1.75%	5741	6.78 ± 0.25	7.05
worst, max 10%	all 43, RMS 3.5%	61579	72.7 ± 2.5	54.9
opposite, max 5%	all 43, signs reversed	999	1.18 ± 0.06	1.40
opposite, max 10%	all 43, signs reversed	4409	5.21 ± 0.20	6.46
worst, top 1 only	M2:`dXIn` -10% (-6.8 cm)	1594	1.88 ± 0.08	1.99
worst, top 3 only	+ M7:B -10%, M7:`Z_len` -8%	5824	6.88 ± 0.25	6.47
worst, top 5 only	+ two yoke widths ($\pm 6.8\%$)	10937	12.9 ± 0.5	11.1
worst, top 10 only	the ten largest changes	29146	34.4 ± 1.2	28.8
single	M7:B -10% ($\lvert B\rvert$ 1.9 $\to$ 1.71 T)	1790	2.11 ± 0.09	2.03
single	M2:`dXIn` -10% (-6.8 cm)	1656	1.96 ± 0.08	1.99
single	M7:`dXOut` -10% (-6.8 cm)	1563	1.85 ± 0.08	1.41
single	M7:`Z_len` -10% (-19.3 cm)	1395	1.65 ± 0.07	1.69
single	M7:`dXYokeIn` $+10\%$ ($+12.5$ cm)	1315	1.55 ± 0.07	1.47
single	M2:B -10% (1.9 $\to$ 1.71 T)	1071	1.26 ± 0.06	1.34
single	M3:`dXYokeIn` -10% (-8.6 cm)	1070	1.26 ± 0.06	1.45
single	M6:`Z_len` -10% (-16.8 cm)	1040	1.23 ± 0.06	1.30

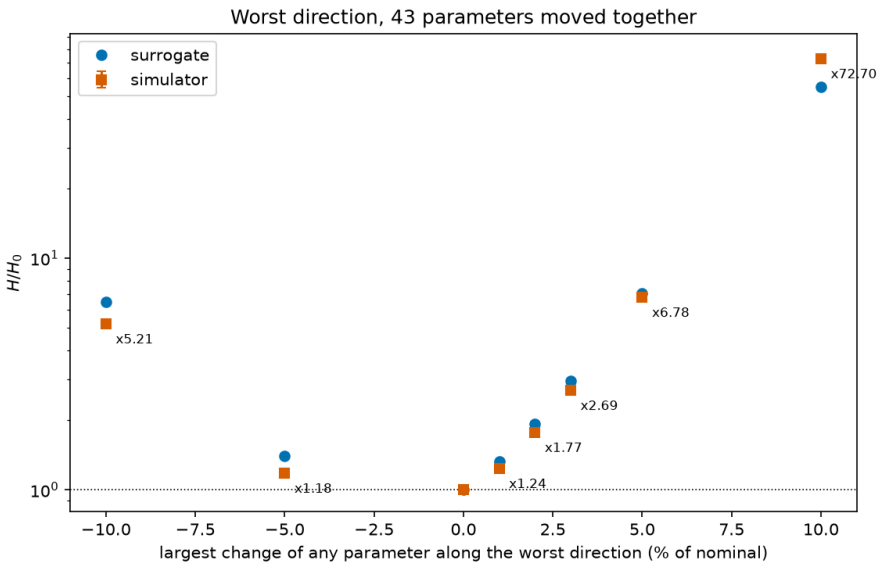


Figure 9: Hits along the worst direction against the largest change of any parameter: simulator and surrogate.

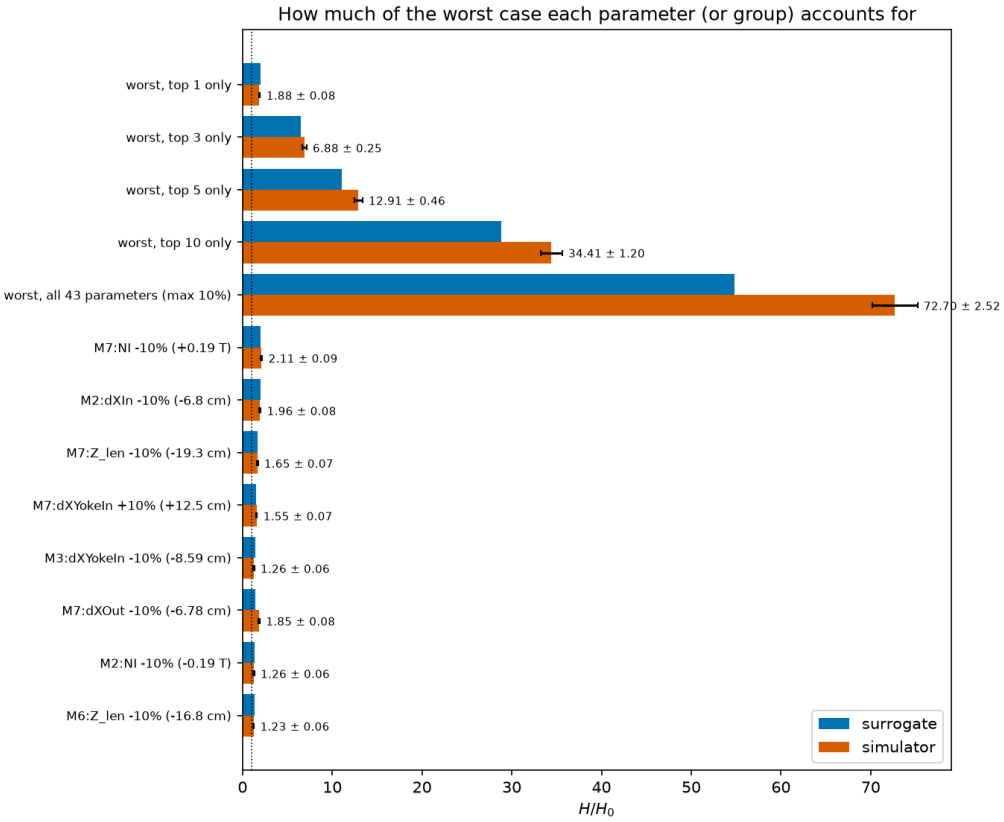


Figure 10: The worst direction at a maximum change of 10%, truncated to its largest 1, 3, 5, 10 changes, and single parameters at their worse limit.

4.5 The worst case for a given tolerance

Method. If every parameter may be off by up to ε, the designs allowed form the box $|u_i| \leq \varepsilon/10\%$. Its worst design is found on the surrogate, by projected-gradient ascent of $\log H$ over the box from 16 starting points (the sign of the gradient, the scaled worst direction and random corners; 300 Adam steps each, all batched, seconds per box), and then simulated on the 10^9 muons of the walks. The simulated values are real designs within the tolerances, so the true worst case is at least as bad; the search is as good as the surrogate, which is accurate there up to $\pm 3\%$ and overestimates beyond. The worst direction of Sec. 4.3 maximises the hits on a sphere ($\|u\| = 1$); on a box the maximum is at a corner, every parameter at a limit with the harmful sign, so scaling the direction down underestimates it.

Table 7: The worst design within $\pm\varepsilon$ on every parameter: found on the surrogate, simulated (10^9 muons; nominal 849 ± 21, two runs). Last column: the worst direction scaled to the same ε (Table 6).

ε	at a limit	sim. hits	sim. H/H_0	surr. H/H_0	scaled direction (sim.)
1%	43	1462	1.72 ± 0.07	1.79 ± 0.04	1.24 ± 0.06
2%	43	3207	3.78 ± 0.15	3.91 ± 0.15	1.77 ± 0.08
3%	43	7644	9.0 ± 0.3	9.5 ± 0.4	2.69 ± 0.11
5%	39	112 445	132 ± 5	236 ± 18	6.78 ± 0.25
10%	39	2 917 029	3440 ± 120	$(1.1 \pm 0.7) \times 10^5$	72.7 ± 2.5

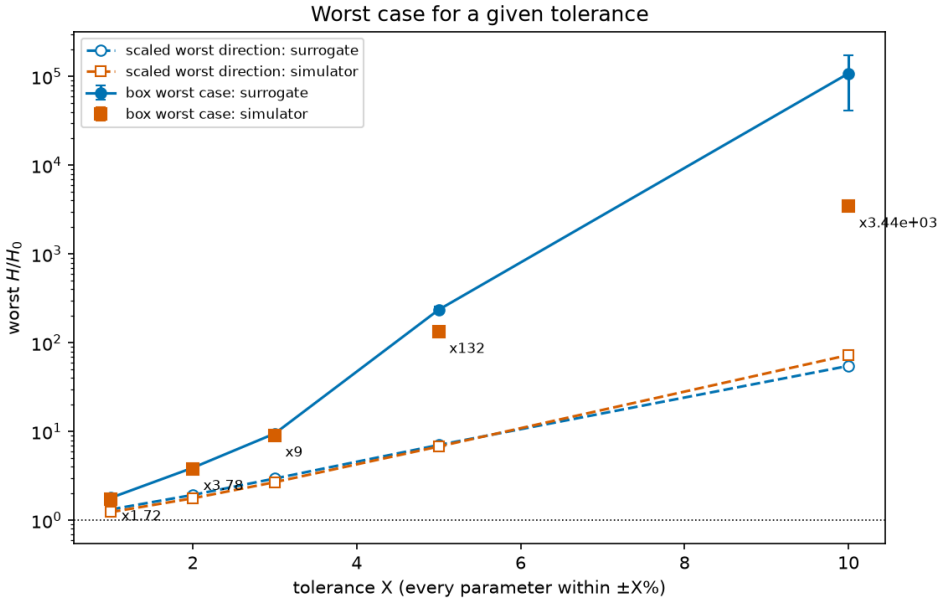


Figure 11: Worst case for a given tolerance: the worst design within $\pm\varepsilon$ (filled; surrogate and simulator) and the worst direction scaled to ε (open).

Results (Table 7, Fig. 11). Already at $\pm 1\%$ the worst design gives $\times 1.7$, at $\pm 2\%$ $\times 3.8$ and at $\pm 3\%$ $\times 9.0$, 1.4 to three times the scaled direction. Up to $\pm 3\%$ these designs act through the known mechanism: the largest contributions are those of the worst direction (M7 weaker and shorter, M2's entrance narrower, M7's exit and M3's inner yoke narrower), but every parameter is at its limit ($\cos \approx 0.62$ with the worst direction), and the surrogate predicts them to 3–5%. At $\pm 5\%$ and $\pm 10\%$ the search finds another mechanism, M7's yokes narrower with M2's entrance and M3's inner yoke wider ($\cos = -0.26$ and -0.12 with the worst direction), which the local derivatives do not show; the simulated scans point the same way (M7's inner yoke harmful on both sides, Fig. 4). There the surrogate overestimates, by $\times 1.8$ at $\pm 5\%$ and $\times 30$ at $\pm 10\%$, but the designs it finds are real: $\times 132$

at ±5% and ×3440 at ±10%. The cruder box search of the walks (`pga`, 4 starts, 200 steps) ends at a corner with $\times 1070 \pm 40$, almost all from muons of 40–60 GeV; halfway to it (every parameter at 5%) the simulator gives $\times 34 \pm 1$, and its opposite corner $\times 28 \pm 1$. Which muons make the hits of the ±5% and ±10% worst designs was not measured.

4.6 Tolerance allocation

Method. Which tolerances keep the worst case below a target? Greedily: start with every parameter at ±10%; find the box's worst design u^* as above; tighten the two parameters that contribute most to it (the largest first-order shares of $\log H$, $u_i^* \, \partial \log H / \partial u_i$) by a factor 0.8, down to a floor of ±0.5%; repeat until the worst case is below the target. The first box that meets each target is its allocation, and its worst design is simulated. This is a heuristic, not an optimum: other allocations may loosen some parameters for the same target.

Table 8: Tolerance allocations (greedy, surrogate) for a worst case of at most 3, 2 and 1.5 times the nominal hits: the sixteen tightest parameters (±%), and a summary of all 43. The worst design of each allocation is simulated in the last row.

parameter	nominal	$\leq 3H_0$	$\leq 2H_0$	$\leq 1.5H_0$
M2:`dXIn`	68.04 cm	0.55	0.50	0.50
M7:B	1.9 T	0.50	0.50	0.50
M2:`dXIn`	68.04 cm	0.55	0.50	0.50
M7:`Z_len`	193.1 cm	0.55	0.50	0.50
M7:`dXYokeIn`	124.8 cm	1.07	0.50	0.50
M7:`dXOut`	67.81 cm	0.69	0.55	0.50
M3:`dXYokeIn`	85.88 cm	1.07	0.55	0.50
M3:B	1.9 T	1.07	0.69	0.50
M6:`Z_len`	168.0 cm	1.34	0.69	0.50
M3:`dXIn`	64.81 cm	1.34	0.86	0.50
M7:`dXYokeOut`	94.92 cm	1.34	0.86	0.50
M4:`dXIn`	39.02 cm	2.10	1.07	0.55
M4:B	1.9 T	2.10	1.07	0.55
M2:`dXYokeIn`	70.62 cm	2.62	1.34	0.55
M7:`dXIn`	37.95 cm	1.68	1.07	0.69
M2:B	1.9 T	2.10	1.34	0.69
M3:`Z_len`	225.0 cm	1.34	0.86	0.86
parameters below ±1% / ±2% / ±5%		4 / 14 / 30	11 / 18 / 36	20 / 27 / 41
median tolerance		5.1%	2.6%	1.1%
worst design: surrogate		2.97 ± 0.26	1.99 ± 0.06	1.50 ± 0.04
worst design: simulator		2.44 ± 0.10	1.56 ± 0.07	1.40 ± 0.06

Results (Table 8, Fig. 12). A guarantee against the worst case is demanding. For at most ×3, the allocation holds M7's field to ±0.5% (0.01 T), M2's entrance width and M7's length to ±0.55% (0.37 and 1.1 cm) and M7's exit width to ±0.7%; 14 parameters are below ±2%, two may stay at ±10%, and the median tolerance is 5%. For at most ×2 it puts eleven parameters below ±1%, and for ×1.5 twenty, ten of them at the 0.5% floor; with every parameter at ±0.5% the worst case is still ×1.29. The order follows the sensitivities: M7 (field, length, exit width, yokes), M2's entrance and M3 (inner yoke, field, entrance) first, the gaps between magnets last. The simulated worst designs stay below every target (×2.44, 1.56, 1.40): at these small tolerances the surrogate is conservative.

4.7 Random tolerances

Designs with every parameter independently uniform in ±2%, ±5% and ±10%: eight of each simulated, and 4096 per level from the surrogate (Fig. 13).

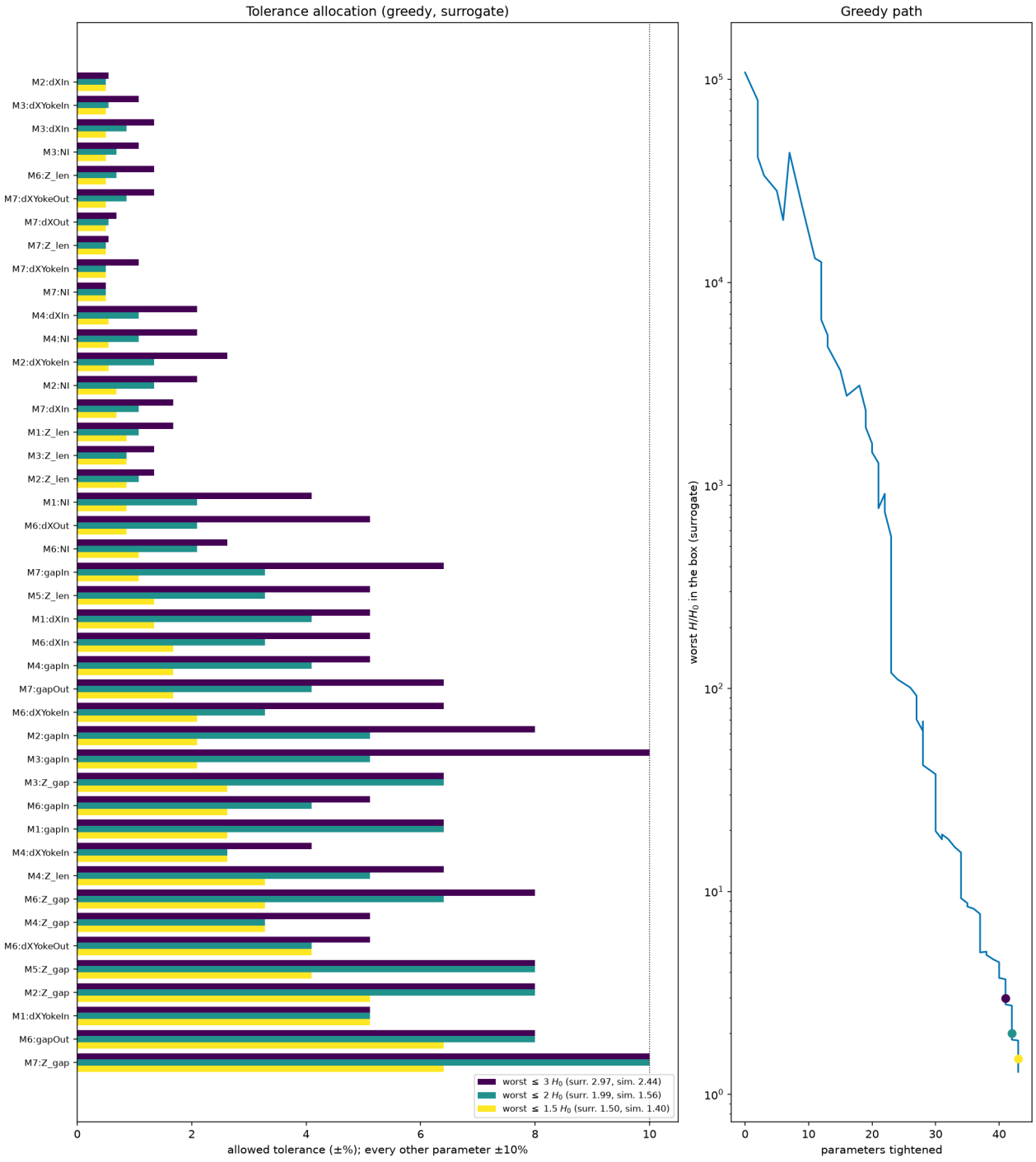


Figure 12: Left: the tolerance allocations for a worst case of at most 3, 2 and 1.5 H_0 (surrogate; simulated worst designs in the legend). Right: the greedy path, the worst case of the box against the number of parameters tightened.

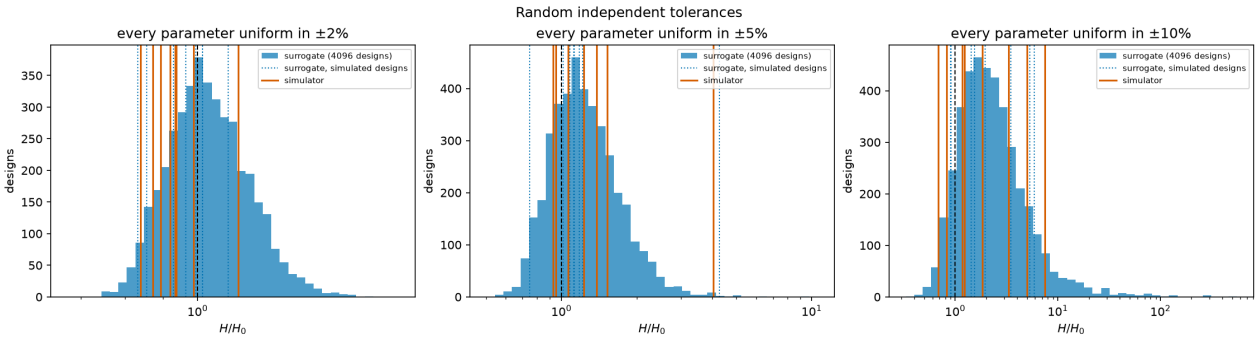


Figure 13: Random independent tolerances: surrogate distribution of 4096 designs per level (histogram), and the eight simulated designs (solid: simulator, dotted: surrogate).

tolerance	simulated H/H_0 (8 designs)	surr. median	surr. 5–95%	$P(H > H_0)$
±2%	0.84, 0.87, 0.89, 0.92, 0.93, 0.94, 0.99, 1.13	1.03	0.85–1.27	58%
±5%	0.93, 0.95, 1.06, 1.07, 1.23, 1.39, 1.52, 4.04	1.20	0.78–2.16	74%
±10%	0.69, 0.83, 1.17, 1.23, 1.84, 3.30, 5.00, 7.43	2.08	0.82–8.98	90%

At ±2% the hits stay within about ±20%. At ±5% most designs stay within ±50%, but one of the eight simulated quadruples the hits. At ±10% the median design is twice as bad and the tail reaches ×7.4 among the simulated designs (×29 at the surrogate's 99th percentile). The surrogate agrees with the simulator design by design (median ratio 0.88–1.02).

4.8 Which muons respond

The muons are split into bins of p_z (and separately p_t and charge), and each bin's hits H_b get their own gradient, one-at-a-time effects and worst direction (Fig. 15). Below 75 GeV the per-bin magnitudes carry the surrogate's larger uncertainty there (Sec. 2.1), so the directions and signs are the robust part; the simulated hits per p_z range (Fig. 14) confirm them.

Three populations. Half of the muons have $p_z < 30$ GeV, but at the nominal design the hits come from two groups (Table 9; simulated shares in parentheses): *intermediate* momenta, 30–75 GeV, with 66% of the hits (66%; 43% at 40–50 GeV alone) at a hit rate of 10^{-6}; and *high* momenta above 125 GeV, 1.2% of the muons but 26% of the hits (24% above 150 GeV), with hit rates up to 10^{-3} above 300 GeV: the muons the shield does not deflect enough. A third group, *soft* muons of 20–30 GeV, gives only 4% of the hits but responds strongly to some changes.

The populations want opposite changes. The worst directions of the 30–45 and 45–75 GeV bins agree (cos = 0.95), as do all bins above 125 GeV (0.89–1.00), but the two groups are anti-aligned (cos = −0.24 to −0.47), and the soft bins are anti-aligned with the intermediate ones too (down to −0.61, Fig. 16). The total gradient, and every harmful direction of Table 4, follows the intermediate group (cos = 0.83–0.94). Simulated (Fig. 14):

- worst direction at 10%: the 40–50 GeV hits go from 363 to 44 637 and the 50–60 GeV ones from 67 to 14 346, while those above 150 GeV *fall* from 202 to 45 and those below 30 GeV from 37 to 6;
- opposite direction: the 40–50 GeV hits fall from 363 to 71, but those of 20–30 GeV rise from 37 to 2353 and those above 150 GeV from 202 to 1286;
- the box corner of the walks: only 30–75 GeV (562 → 903 046), the high-momentum hits fall (202 → 42);

Table 9: Per p_z bin at the nominal design (surrogate): share of the muons and of the hits, hit rate, relative sensitivity $|\nabla H_b|/H_b$ per tolerance unit, and the cosine between the bin's gradient and that of the total.

p_z [GeV]	muons	share of hits	hit rate	$\lvert\nabla H_b\rvert/H_b$	$\cos(g_b, g)$
10–20	1.7%	0.0%	1.2×10^{-8}	1.79	−0.17
20–30	48.4%	3.9%	6.6×10^{-8}	2.44	−0.42
30–45	28.8%	42.6%	1.2×10^{-6}	1.65	+0.94
45–75	15.3%	23.3%	1.2×10^{-6}	2.13	+0.83
75–125	4.7%	4.2%	7.2×10^{-7}	0.38	+0.40
125–175	0.89%	2.7%	2.5×10^{-6}	0.43	+0.08
175–225	0.23%	3.0%	1.1×10^{-5}	1.28	+0.07
225–300	0.08%	10.5%	1.1×10^{-4}	1.23	+0.07
> 300	0.006%	9.7%	1.3×10^{-3}	2.96	+0.06

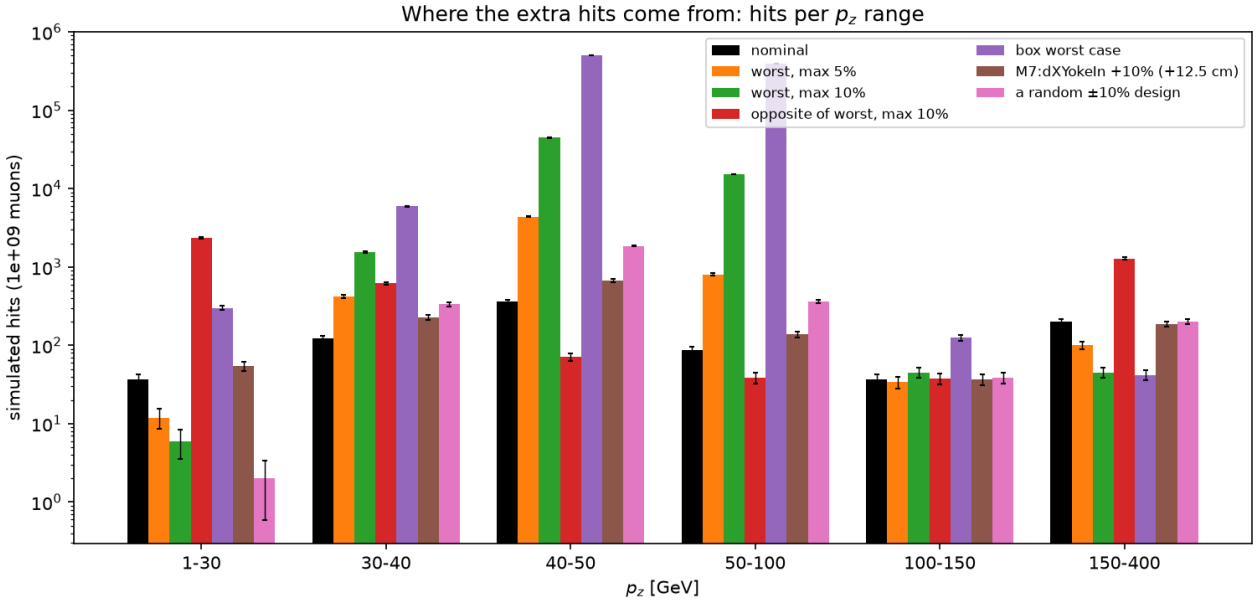


Figure 14: Simulated hits per p_z range for the nominal design and some designs of Table 6.

- random ±10% designs can open any group: one raises the 40–50 GeV hits from 363 to 3060, another those above 150 GeV from 202 to 1033, and the worst of the eight (×7.4) both the soft (37 → 1967) and the high-momentum (202 → 3296) ones.

Which parameters act on which muons (Fig. 17). The high-momentum hits are controlled by the *upstream* magnets: the lengths and fields of M1–M4, each changing the hits above 175 GeV by 40–200% at its worse limit. The intermediate hits are controlled by the entrance width of M2, the yokes of M3 and M7, and above all the *downstream* magnet M7 (its field: +150–170% of the 30–75 GeV hits at the worse limit). The soft hits respond most to the entrance widths of M2 and M3 and the yoke of M3 (up to ×7.8). This explains `act1`: changing M1–M4 together shifts hits between the high-momentum group and the others, and the nominal design balances them.

Transverse momentum and charge. All p_t bins below 1.5 GeV behave alike (worst directions at cos = 0.89–1.00). The muons above 1.5 GeV (1.6% of the muons, 23% of the hits) are less sensitive (0.85 vs. 1.1–1.4 per tolerance unit) and only partly aligned (cos = 0.66–0.72, Fig. 18). Positive muons give 64% of the hits; both charges respond to the same changes (cos = 0.98), the μ^- somewhat more strongly (1.22 vs. 0.99).

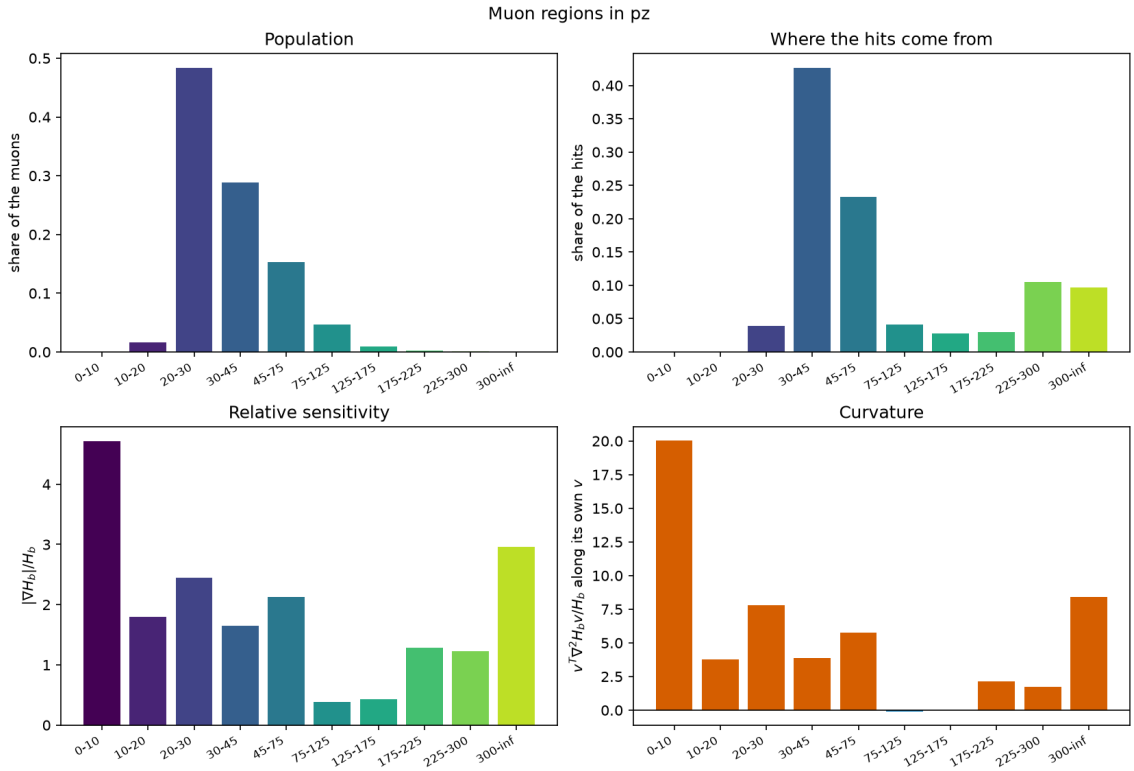


Figure 15: Per p_z bin (surrogate): share of the muons and of the hits, relative sensitivity and curvature along the bin's own worst direction.

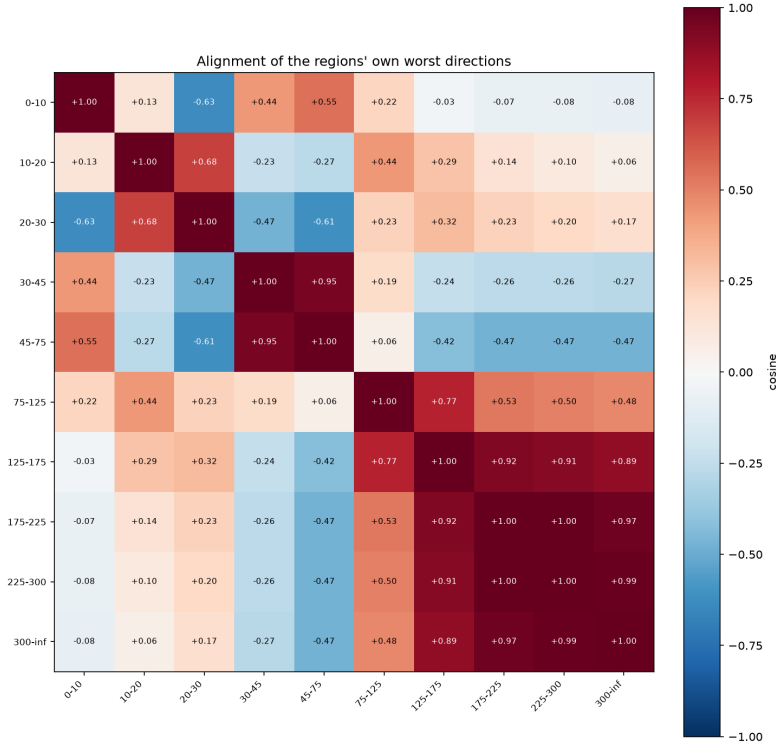


Figure 16: Cosine between the worst directions of the p_z bins: the intermediate momenta (30–75 GeV) want the opposite changes to the high (> 125 GeV) and the soft (10–30 GeV) ones.

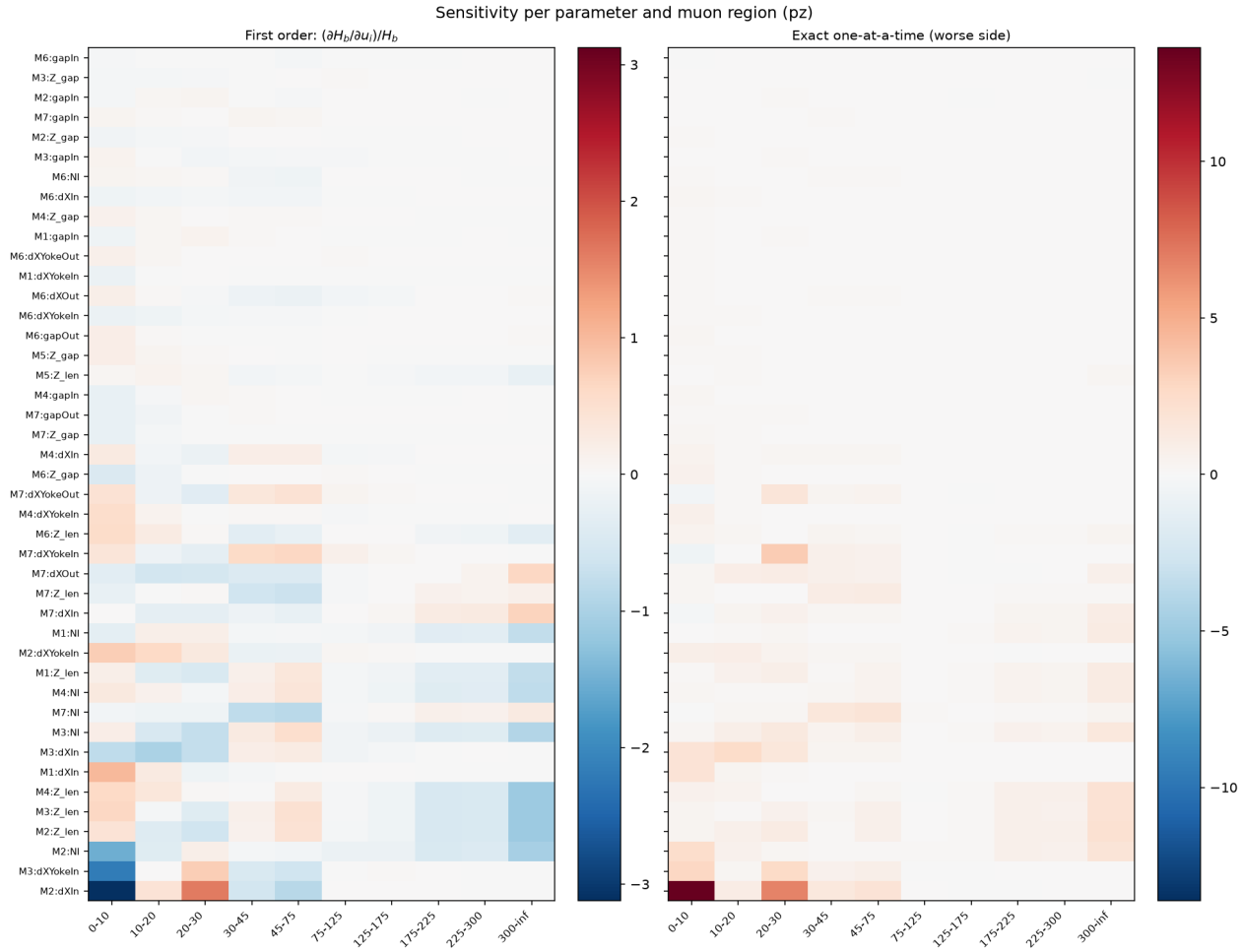


Figure 17: Sensitivity of each p_z bin to each parameter, relative to the bin's own hits. Left: first order. Right: exact one-at-a-time change at the worse of ±10%.

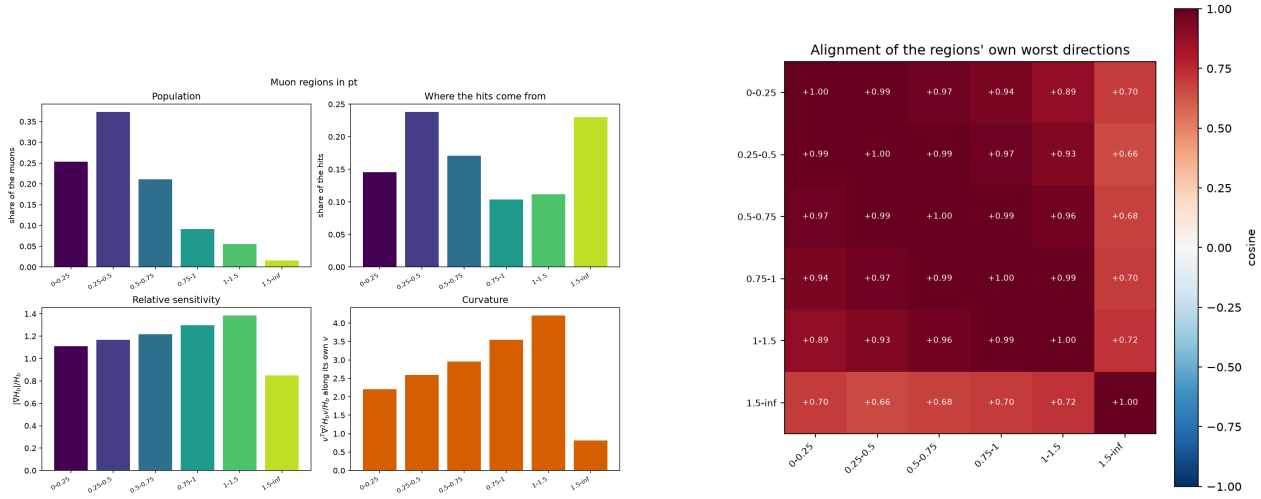


Figure 18: Per p_t bin: overview (left) and alignment of the bins' worst directions (right).

5 Conclusions

1. **The design is fragile to coordinated deviations.** With every parameter off by up to ε, the worst design found (on the surrogate, then simulated) gives $\times 1.7$ at $\varepsilon = 1\%$, $\times 3.8$ at 2%, $\times 9.0$ at 3%, $\times 132$ at 5% and $\times 3440$ at 10%; these are lower bounds on the true worst case, tight up to 3%. Moving every parameter by 1.5% with the sign of its gradient already doubles the hits.

2. **Bounding the worst case needs sub-percent tolerances on a few parameters and a few percent on most.** The allocation found for at most $\times 3$ holds M7's field to $\pm 0.5\%$, M2's entrance width and M7's length to $\pm 0.55\%$ and M7's exit width to $\pm 0.7\%$, 14 parameters below $\pm 2\%$, median 5%; for at most $\times 2$, eleven below $\pm 1\%$ (simulated $\times 2.44$ and $\times 1.56$). M7, M2 and M3 need the tightest tolerances, the gaps between magnets the loosest.

3. **Single parameters underestimate the risk.** The worst one at its $\pm 10\%$ limit (the field of M7) gives $\times 2.1$; the entrance width of M2 and the exit width and length of M7 give $\times 1.65$–2.0. The top three changes of the worst direction together give $\times 7$, the top ten $\times 34$: the factors multiply. M7 matters most, then M2, M3 and M6.

4. **Random deviations are much milder than the worst case.** Independent errors of $\pm 2\%$ keep the hits within about $\pm 20\%$; at $\pm 5\%$ most designs stay within $\pm 50\%$ but some quadruple the hits; at $\pm 10\%$ the median design is twice as bad and the tail reaches $\times 7$–29. A specification can therefore be much looser if the errors are independent than if it must hold against the worst combination.

5. **Different muons pull in opposite directions.** The hits come from muons of 30–75 GeV (66%) and above 150 GeV (24%). Every harmful direction near the nominal design acts through the first group, while the high-momentum hits fall; the reverse direction raises the soft and high-momentum hits instead. The dominant active-subspace direction (the lengths and fields of M1–M4) trades hits between the groups and is harmful on both sides.

6. **The surrogate is quantitative within the tolerance region.** It reproduces the one-at-a-time effects to 4%, the walks within one tolerance unit to 1–2σ and the worst designs within ± 1–3% to 3–5%. It overestimates the worst designs at $\pm 5\%$ ($\times 1.8$) and $\pm 10\%$ ($\times 30$) and is $\sim 13\%$ uncertain far from the nominal design, so every number used for a decision here is simulated.

Limitations

- Uniform fields per magnet block, no field maps.
- Muons come from the generative flow, so the absolute numbers depend on how well the flow reproduces the muon spectrum.
- The surrogate's errors are the scatter of its members: they miss a bias the members share, such as the one far from the nominal design (Sec. 2.1). Only the top Hessian eigenvector is individually determined.
- The random-tolerance distributions rest on eight simulated designs per level and on the surrogate for their tails.
- The worst designs within a tolerance are found on the surrogate: their simulated values bound the true worst case from below, tightly up to $\pm 3\%$, possibly loosely at $\pm 5\%$ and $\pm 10\%$, where the surrogate is unreliable. The tolerance allocation is a greedy heuristic, not an optimum.

A Free parameters

Table 10: Free parameters and nominal values (lengths in cm, B in T).

magnet	Z_gap	Z_len	dXIn	dXOut	gapIn	gapOut	dXYokeIn	dXYokeOut	B
M1	–	151.5	63.12	–	8.149	–	64.12	–	1.9
M2	15.0	224.9	68.04	–	8.431	–	70.62	–	1.9
M3	16.12	225.0	64.81	–	8.986	–	85.88	–	1.9
M4	15.0	224.0	39.02	–	8.327	–	97.53	–	1.9
M5	15.0	120.4	–	–	–	–	–	–	–
M6	15.0	168.0	20.0	30.0	24.92	9.388	97.18	102.7	−1.2
M7	15.0	193.1	37.95	67.81	8.000	8.000	124.8	94.92	−1.9

B Integrating the surrogate over the muons

With $A(z)$ the kinematic envelope of the sampled muons and $N = 10^9$,

$$H(u) = \frac{N}{\int \mathrm{d}^2 z\, \mathcal{N}(z) A(z)} \sum_{q=\pm 1} P(q) \int \mathrm{d}^2 z\, \mathcal{N}(z)\, A(z) \int_0^{2\pi} \frac{\mathrm{d}\varphi}{2\pi}\, p\big(\text{hit} \mid u, x(z, \varphi, q)\big), \tag{3}$$

where z is the flow's 2-d momentum latent, φ the azimuth (uniform) and q the charge (with the data's μ^+ fraction). The derivatives use 801^2 latent points on $[-7, 7]^2$ with Gaussian weights, 64 azimuths and both charges; everything else $201^2 \times 64 \times 2$ points, with the trunk features cached. The grid is converged to 10^{-5} from about 10^7 points (3×10^{-4} at 10^6), while 10^9 sampled muons scatter by up to 2% in H_0 and 5% above 300 GeV, and quasi-Monte Carlo helps only by a factor two (Fig. 19). Importance-sampling the simulator at a raised temperature does not help either: the hits that matter are at intermediate momenta, of which a hotter sample holds fewer.

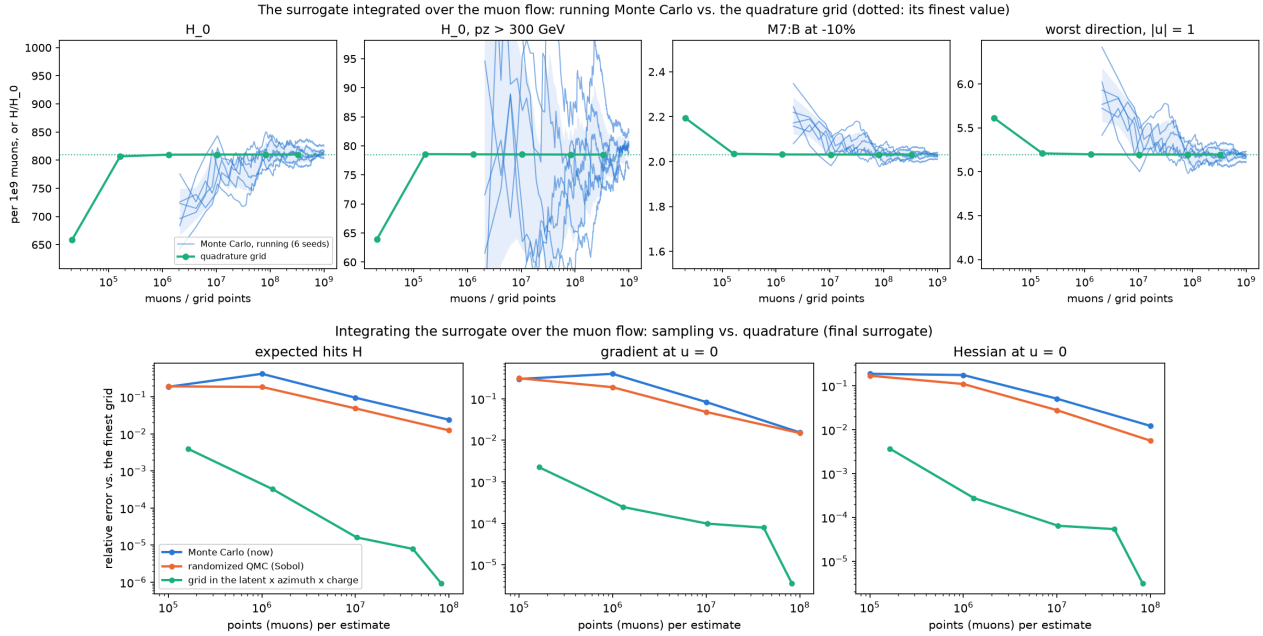


Figure 19: Top: Monte Carlo estimates (six seeds) against the grid at increasing resolution, for the nominal hits, the hits above 300 GeV, the field of M7 at −10% and one tolerance unit along the worst direction. Bottom: relative error against the finest grid at equal numbers of points, for Monte Carlo, randomised quasi-Monte Carlo (Sobol) and the grid.

C How the surrogate was chosen

The choices come from a study of the surrogates (`surrogate_study/`) on four toy problems with exact answers (a binary outcome per sample with known probability; 10–43 parameters) and on the shield against the reference designs (Fig. 20).

- **Modelling each muon is what makes it work.** A response surface fitted to the same simulations without the muons needs about ten times more simulation for the same gradient.
- **Training data.** Many designs with few muons each beat few designs with many; all designs at once beat a rolling buffer; the designs must cover the region where the derivatives are read. A hotter momentum temperature ($T = 2$) is much better than $T = 1$ or 1.5, and $T = 3$ no better than $T = 2$. A tailored training proposal (the provably best one for the error on H) would give at most 1.7× the effective training muons of $T = 2$, so the temperature is kept. Adding designs far from the nominal one helped only with the adaptive pool: the uniform pool alone did not lower the box error on the nominal muons, with the adaptive pool it fell by a fifth, to 13% (four seeds each, scored on the same references).
- **Architecture.** The DeepONet is the most reliable for the hit rate. At the final scale (2×10^8 rows) capacity matters (128 units, 32 features), and averaging seeds removes most of the scatter of the gradient. Reduced Taylor surrogates (a per-muon quadratic on a low-dimensional subspace), the best Hessians on the toys, are worse here (one-at-a-time secants 6–10%, curvatures 18–25%). An MLP on $[u, x]$ predicts somewhat better (box 11% instead of 16%, curvatures 10% instead of 12%) and agrees with the DeepONet's gradient, top eigenvector and worst direction to cos = 0.99, so the conclusions do not depend on the choice; the DeepONet is kept because its separable form allows cached trunk features and exact derivatives.

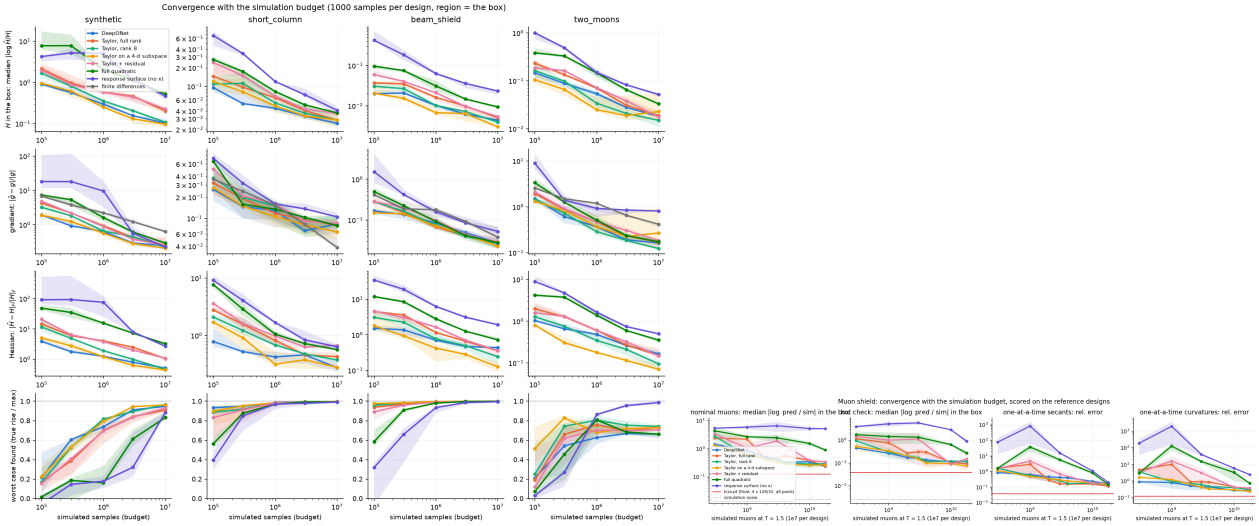


Figure 20: Left: toy problems, errors of the hit rate, gradient and Hessian, and the share of the true worst case found, against the simulation budget, per architecture. Right: shield, errors on the reference designs against the budget (single models); solid red: the final surrogate, dotted: simulation noise.